

REMUS WONG

SOFTWARE ENGINEER

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EDUCATION

The University of Texas at Austin	Bachelor of Science in Computer Science Concentration in Video Game Development	Spring 2022
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SKILLS

Languages: Typescript/Javascript, C++, Python, C#, Java, Rust

Frameworks: ThreeJS, BabylonJS, Qt, React, Flask

Graphics: WebGPU, WebGL, GLSL, WGSL, Slang, OpenGL

Tools: Unity, Unreal Engine, Git, Houdini, Docker

Database: AWS (RDS, S3), PostgreSQL

RELEVANT EXPERIENCE

CLO Virtual Fashion — *Web Graphics Engineer* January 2023—Present

- Contributed to improving the visual fidelity of 3D garments on the web using Three.js/Babylon.js
- Optimized the order-independent transparency algorithm (depth peeling), increasing FPS by 15%
- Analyzed 3D data download speeds and optimized the strategy, improving performance by 22%
- Researched and developed a custom C++ and WebGPU render engine for a next-generation cross-platform renderer
- Designed a data-oriented approach for loading custom 3D garment data for the new WebGPU render engine
- Integrated and tested various order-independent transparency algorithms (Weighted-blended and Spinlock)
- Implemented supersampling anti-aliasing to enhance image quality
- Enhanced benchmark tests to profile and log GPU usage, enabling better tracking of performance changes
- Collaborated with an international team to directly resolve issues with users and plan user oriented goals

Aspyr Media — *Associate Technical Artist* July 2022—January 2023

- Initiated the implementation of a NoSQL database to store assets for a content management system
- Refactored the tools launcher for modular deployment across users

Corvid Technologies — *Virtual Reality Intern* May 2021—August 2021

- Streamlined preprocessing of high-fidelity CAD models for integration into an Unreal Engine VR application
- Enhanced pipeline tools using PyQt and Unreal Python API to automate import/export of custom levels for Unreal
- Created an AR experience for interacting with real-time vehicular airflow simulations using HoloLens 2

PERSONAL PROJECTS

Ready Recipes — *Full Stack Developer*

- Designed front-end with React to serve users information about regional food recipes
- Modeled a PostgreSQL database using AWS RDS to store relationships between recipes, ingredients, and location
- Created RESTful API using Flask and Postman to facilitate backend services

2D Unity Game Capstone — *Programmer / Game Designer*

- Programmed multiple game systems like dialogue system and enemy AI as a principal programmer for the team
- Applied agile methodology over the course of the semester in a team of six to coordinate consistent progress
- Led development of key game design decisions through technical documentation and communication

OpenGL Raytracer — *Programmer*

- Implemented Whitted illumination model to ray trace 3D scenes
- Added support for triangular meshes with triangle-ray intersection and Phong interpolation for per-vertex normals
- Added bilinear texture mapping for models and custom cube maps for scene customization